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Midterm I Part C
Electromagnetic Theory
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Q1: Consider a spherical shell as shown in Fig. 1(a) with inner radius a and outer radius b. Let there be filled two types of dielectric material with relative permittivities ε_{r1} and ε_{r2}, forming two regions, 1 and 2, respectively. Let $a = 1$ mm, $b = 4$ mm, $\varepsilon_{r1} = 2$, and $\varepsilon_{r2} = 4$. Assume that the charge distribution is uniform all over the inner and outer spheres.

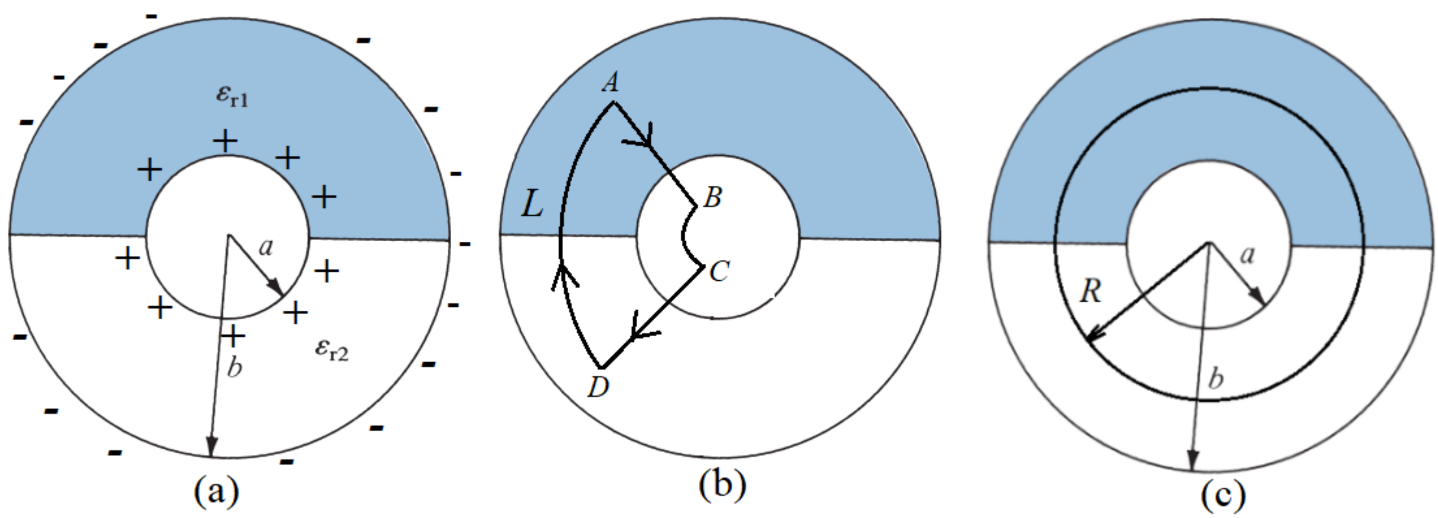

Figure 1: Illustrations for **Q1**.

(a) Sketch the electric flux lines.

(b) Consider a closed loop path L (connecting points A, B, C and D) as shown in Fig. 1(b) below:

Evaluate the integral

$$\oint_L \mathbf{E} \cdot d\mathbf{l} = 0$$

What relation do you find in the electric field intensities of regions 1 and 2?

(c) Based on part (b), what relation you may establish between electric flux densities of regions 1 and 2?

(d) Consider a closed *Gauss'* surface of an arbitrary radius R such that $a < R < b$ as shown in Fig. 1(c).

Evaluate the integral

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q$$

What relation do you find in the electric flux densities of the two regions and the charge $+Q$?

(e) Based on above findings, write down the expressions of $\mathbf{E}_1$, $\mathbf{E}_2$, $\mathbf{D}_1$ and $\mathbf{D}_2$.

(f) Find the potential of the inner sphere with respect to the outer sphere by solving the following integral where the value of E is what you obtained in part (e):

$$V_{ab} = -\int_b^a \mathbf{E} \cdot d\boldsymbol{\ell}$$

(g) Based on the findings of part (f), find the total (equivalent) capacitance C_{eq}. Also find the capacitances C_1 and C_2 of regions 1 and 2, respectively. What is the relation among C_{eq}, C_1 and C_2.

(h) Find the total energy stored in the regions between spherical plates of the capacitor. Solve the following integral:

$$W_E = \frac{1}{2} \int_V \mathbf{D} \cdot \mathbf{E}\, dv$$

Hint: You may apply this formula separately for regions 1 and 2, and then add them together to compute total energy.

(i) Using the findings of part (h), how can you find the total (equivalent) capacitance C_{eq}?

Hint: May we use the following expression?

$$W_E = \frac{1}{2} C_{\text{eq}} V_{ab}^2.$$

Q2: Consider a piece of a concentric cylindrical cable. There is a hollow in the middle. A material is filled between the inner and outer conducting circular sheets at radius a and b, respectively, where $b > a$. Refer to Fig. 2 for illustration.

Assume that the potential of the conducting sheet at $r = a$ is V_0, the potential of the conducting sheet at $r = b$ is zero, and L is the length of each conducting circular sheet. The conducting sheets have negligible resistances, so the resistance is offered by the material filled between the sheets. The conductivity of the material is

$$\sigma = \frac{A}{r} + B$$

where A and B are constants, and r is the radius such that $a < r < b$.

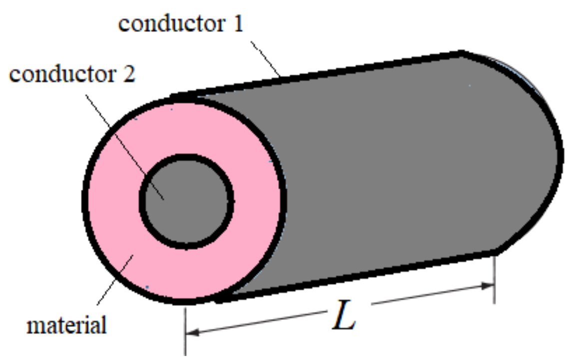

Figure 2: Illustration for **Q2**.

(a) Show that the current density (for length L) is given by

$$\mathbf{J} = \frac{I}{2\pi r L}\widehat{\mathbf{r}}$$

where I is the current.

(b) Find the electric field intensity in the medium.
Hint: Exploit the relation between $\mathbf{J}$ and $\mathbf{E}$ for the simple conducting medium.

(c) Using the electric field $\mathbf{E}$, establish the relation between V_0 and current I by solving the integral:

$$V_0 = V_{ab} = -\int_b^a \mathbf{E} \cdot d\boldsymbol{\ell}$$

and finally find the expression of resistance from $R = V_0/I$.

Answers of Q1:

$$\mathbf{D}_1 = \frac{Q\,\epsilon_1\,\widehat{\mathbf{r}}}{2\pi r^2\,(\epsilon_1+\epsilon_2)}, \quad \mathbf{D}_2 = \frac{\epsilon_2}{\epsilon_1}\mathbf{D}_1 \quad \text{and} \quad \mathbf{E}_1 = \mathbf{E}_2 = \frac{Q\,\widehat{\mathbf{r}}}{2\pi r^2\,(\epsilon_1+\epsilon_2)}$$

Hence, the capacitance of the system is

$$C_{\text{eq}} = 2\pi\,(\epsilon_1+\epsilon_2)\left(\frac{ab}{b-a}\right) = C_1 + C_2$$

where $C_1 = 2\pi\epsilon_1\left(\frac{ab}{b-a}\right)$ and $C_2 = 2\pi\epsilon_2\left(\frac{ab}{b-a}\right)$.

Note that C_1 and C_2 are the capacitances of medium 1 and medium 2, respectively. Thus, the capacitance of the system is equivalent to the parallel combination of the two capacitances. You may have already used this result in the analysis of electrical circuits in **EE-111** - Electric Circuit Analysis.

Answers of Q2:

$$R = \frac{M}{2\pi LB}, \text{ where } M = \ln\left(\frac{A+bB}{A+aB}\right)$$